A Degree-Two Default Mechanism for Single-Nomination Impartial Selection

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Abstract

We study randomized impartial selection in the single-nomination model, where each vertex of a directed graph has outdegree one and the objective is to approximate the maximum indegree. Cembrano, Fischer, and Klimm proved a general lower bound of 2105/3147 and upper bounds tending to 3/4, leaving open whether asymptotic 3/4-optimality is achievable. This note gives a small but rigorous improvement in two directions. First, we construct a simple impartial mechanism, the degree-two default mechanism D_2, and prove that on every n-vertex graph with maximum indegree 2 it achieves

$$\frac{3}{4} - \frac{3}{4n}.$$

Thus the known asymptotic upper-bound family based on maximum indegree 2 is matched up to $O(1/n)$ by an explicit mechanism on that degree class. Second, by mixing D_2 with the permutation mechanism and PRUGD of Cembrano, Fischer, and Klimm, we obtain an asymptotic worst-case guarantee

$$\frac{12421}{18558} = 0.669307\ldots,$$

which improves the previous asymptotic lower bound $2105/3147 = 0.668891\ldots$. The improvement is modest and does not resolve the 3/4 problem, but it isolates a useful impartial primitive targeted at the degree-two obstruction.

1 Model and notation

Let $G = (V, E)$ be a directed graph with no self-loops and exactly one outgoing edge from every vertex. We write $|V| = n$ and

$$d_G(v) = \delta^-(v, G) = |\{u \in V : (u, v) \in E\}|$$

for the indegree of v. The maximum indegree is

$$\Delta(G) = \max_{v \in V} d_G(v).$$

For a vertex v, let $G - v$ denote the partial graph obtained from G by deleting the outgoing edge of v. More generally, if H is a partial graph in which some outgoing edges may already be absent, $H - v$ denotes the partial graph obtained by deleting the outgoing edge of v if it is present.

A randomized selection mechanism assigns to each graph G a probability distribution over V. It is impartial if the probability of selecting v is unchanged by changing the outgoing edge of v. The performance ratio of a mechanism F on G is

$$\frac{\mathbb{E}[d_G(F(G))]}{\Delta(G)}$$

when $\Delta(G) > 0$.

For a partial graph H, define

$$B_2(H) = \{v \in V : d_H(v) = 2\}, \qquad b_2(H) = |B_2(H)|.$$

2 The degree-two inexact rule

We first define an inexact rule. It may select no vertex with positive remaining probability; the remaining probability will later be assigned to a default vertex.

Definition 2.1 (The inexact rule q_2)**.** For a partial graph H, define

$$q_{2,v}(H) = \begin{cases} \dfrac{1}{2\, b_2(H - v)}, & \text{if } d_H(v) = 2, \\ 0, & \text{otherwise.} \end{cases}$$

This is well-defined: if $d_H(v) = 2$, then deleting the outgoing edge of v does not change the indegree of v, so $b_2(H - v) \geq 1$.

Lemma 2.2. *For every partial graph H,*

$$\sum_{v \in V} q_{2,v}(H) \leq 1.$$

Moreover, q_2 is impartial as an inexact rule: $q_{2,v}(H)$ depends only on $H - v$.

Proof. The impartiality assertion is immediate from the definition, because $d_H(v) = d_{H-v}(v)$ and the denominator is $b_2(H - v)$.

Let $k = b_2(H)$. If $k = 0$, the sum is zero. If $k = 1$, the unique vertex v with indegree 2 contributes at most $1/2$, because $b_2(H - v) \geq 1$. If $k \geq 2$, then deleting one outgoing edge can destroy at most one indegree-two vertex, so for every $v \in B_2(H)$,

$$b_2(H - v) \geq k - 1.$$

Therefore

$$\sum_{v \in V} q_{2,v}(H) = \sum_{v \in B_2(H)} \frac{1}{2b_2(H - v)} \leq \frac{k}{2(k - 1)} \leq 1.$$

□

Definition 2.3 (Degree-two default mechanism D_2)**.** On input $G = (V, E)$, choose a default vertex $r \in V$ uniformly at random and form $H = G - r$. Select a vertex v with probability $q_{2,v}(H)$. With the remaining probability

$$1 - \sum_{v \in V} q_{2,v}(H),$$

select the default vertex r.

Lemma 2.4. *The mechanism D_2 is impartial.*

Proof. Fix a vertex x. Its selection probability under D_2 is

$$\frac{1}{n}\left(q_{2,x}(G-x)+1-\sum_{v\in V} q_{2,v}(G-x)\right)+\frac{1}{n}\sum_{r\in V\setminus\{x\}} q_{2,x}(G-r).$$

For $r \neq x$, the term $q_{2,x}(G-r)$ depends only on $(G-r)-x$, and hence not on the outgoing edge of x. The terms $q_{2,x}(G-x)$ and $\sum_v q_{2,v}(G-x)$ are also determined after the outgoing edge of x has been removed. Thus changing the outgoing edge of x does not change the displayed probability. □

We will repeatedly use the following elementary observation.

Lemma 2.5. *Suppose H has maximum indegree at most 2 and $b_2(H) \geq 1$. Then*

$$\sum_{v\in V} q_{2,v}(H) \geq \frac{1}{2}.$$

Proof. Let $k = b_2(H) \geq 1$. Since H has no vertex of indegree larger than 2, deleting an outgoing edge cannot create a new indegree-two vertex. Hence $b_2(H-v) \leq k$ for every $v \in B_2(H)$. Therefore

$$\sum_{v\in V} q_{2,v}(H) = \sum_{v\in B_2(H)} \frac{1}{2b_2(H-v)} \geq k \cdot \frac{1}{2k} = \frac{1}{2}.$$

□

3 Graphs of maximum indegree two

The next theorem is the main structural guarantee for D_2.

Theorem 3.1. *For every n-vertex single-nomination graph G with $\Delta(G) = 2$,*

$$\frac{\mathbb{E}[d_G(D_2(G))]}{2} \geq \frac{3}{4} - \frac{3}{4n}.$$

If G has at least two vertices of indegree 2, then the stronger bound

$$\frac{\mathbb{E}[d_G(D_2(G))]}{2} \geq \frac{3}{4}$$

holds.

Proof. For a default vertex r, put $H_r = G - r$ and

$$Q_r = \sum_{v\in V} q_{2,v}(H_r).$$

When $\Delta(G) = 2$, every vertex selected by q_2 has full indegree 2 in G. Conditional on default vertex r, the expected selected indegree is therefore

$$X_r = 2Q_r + (1-Q_r)d_G(r) = d_G(r) + Q_r(2 - d_G(r)).$$

First suppose that G has at least two vertices of indegree 2. Deleting one outgoing edge can destroy at most one of them, so H_r has at least one indegree-two vertex for every r. By Lemma 2.5, $Q_r \geq 1/2$. Since $d_G(r) \leq 2$,

$$X_r \geq d_G(r) + \frac{1}{2}(2 - d_G(r)) = 1 + \frac{1}{2}d_G(r).$$

Averaging over the uniformly random default vertex and using $\sum_r d_G(r) = n$ gives

$$\mathbb{E}[d_G(D_2(G))] = \frac{1}{n}\sum_r X_r \geq 1 + \frac{1}{2n}\sum_r d_G(r) = \frac{3}{2}.$$

This proves the ratio 3/4 in this case.

Now suppose that G has a unique vertex x of indegree 2. Since the total indegree is n, there is also a unique vertex of indegree 0, and all remaining vertices have indegree 1. Let

$$N^-(x) = \{a, b\}.$$

If $r \notin \{a, b\}$, then H_r still contains the indegree-two vertex x, so the preceding argument gives $X_r \geq 1 + d_G(r)/2$. If $r \in \{a, b\}$, we use only the trivial lower bound $X_r \geq d_G(r)$, which is exact when H_r has no indegree-two vertex.

Let $s = d_G(a) + d_G(b)$. The vertices a and b are not equal to x, and among vertices other than x there is only one vertex of indegree 0; hence $s \geq 1$. Therefore

$$\begin{aligned}
\mathbb{E}[d_G(D_2(G))] &\geq \frac{1}{n}\left(\sum_{r\notin\{a,b\}}\left(1 + \frac{1}{2}d_G(r)\right) + d_G(a) + d_G(b)\right) \\
&= \frac{1}{n}\left(n - 2 + \frac{1}{2}(n - s) + s\right) \\
&= \frac{3}{2} - \frac{2}{n} + \frac{s}{2n} \geq \frac{3}{2} - \frac{3}{2n}.
\end{aligned}$$

Dividing by $\Delta(G) = 2$ yields the claimed bound. □

4 Auxiliary bounds for maximum indegree three

The degree-two mechanism is not designed for all graphs, but two simple lower bounds for $\Delta = 3$ are useful when it is used with small mixing probability.

Lemma 4.1. *Let G be an n-vertex graph with a unique vertex x of indegree 3 and no vertex of indegree 2. Then*

$$\mathbb{E}[d_G(D_2(G))] \geq 1 + \frac{3}{n}.$$

In particular, D_2 has ratio at least 1/3 on this class.

Proof. If the default vertex r is not an in-neighbor of x, then deleting the outgoing edge of r creates no indegree-two vertex, and D_2 selects the default vertex. If $r \in N^-(x)$, then x has indegree 2 in $G - r$ and is selected by q_2 with probability 1/2; in the original graph its indegree is 3. Thus the gain over selecting the default vertex is

$$\frac{1}{2}(3 - d_G(r)).$$

The three in-neighbors of x are distinct from x and have indegree at most 1, so each gain is at least 1. Since a uniformly random default vertex has expected indegree 1, the result follows. $\square$

Lemma 4.2. *On every graph G with $\Delta(G) = 3$,*

$$\mathbb{E}[d_G(D_2(G))] \geq \frac{2}{3}.$$

Consequently, D_2 has ratio at least $2/9$ on every graph with $\Delta(G) = 3$.

Proof. For a fixed default vertex r, every non-default vertex selected by q_2 has indegree at least 2 in the original graph. Hence the conditional expected indegree is at least

$$\min\{d_G(r), 2\}.$$

Let $c_i = |\{v : d_G(v) = i\}|$ for $i = 0, 1, 2, 3$. Since the total indegree is n,

$$c_1 + 2c_2 + 3c_3 = n.$$

The average of $\min\{d_G(r), 2\}$ over a uniformly random default vertex is

$$\frac{c_1 + 2c_2 + 2c_3}{n} = 1 - \frac{c_3}{n}.$$

Also

$$c_0 = c_2 + 2c_3,$$

because subtracting the number of vertices from the total indegree gives

$$-c_0 + c_2 + 2c_3 = 0.$$

Thus $n = c_0 + c_1 + c_2 + c_3 \geq 3c_3$, and so $1 - c_3/n \geq 2/3$. $\square$

5 Mixing with known mechanisms

We use three known facts from Cembrano, Fischer, and Klimm [1]. Let Perm be the permutation mechanism, and let PRUGD be their default-vertex version of plurality with runner-up and gap.

Proposition 5.1 (Cembrano–Fischer–Klimm)**.** *For $n \geq 6$, the following guarantees hold.*

(i) *If $\Delta = 2$, then* Perm *has ratio at least* $2/3$*, and* PRUGD *has ratio at least* $65/96$*.*

(ii) *If $\Delta = 3$ and there is a unique vertex of indegree at least* 2*, then* Perm *has ratio at least* $2/3$*, and* PRUGD *has ratio at least* $13/18$*.*

(iii) *If $\Delta = 3$ and at least two vertices have indegree at least* 2*, then* Perm *has ratio at least* $31/45$*, and* PRUGD *has ratio at least* $25/42$*.*

(iv) *If $\Delta \geq 4$, then* Perm *has ratio at least*

$$\alpha_1(\Delta) = \begin{cases} \dfrac{3\Delta + 2}{4\Delta + 4}, & \Delta \textit{ even}, \\ \alpha_1(\Delta - 1), & \Delta \textit{ odd}, \end{cases}$$

and PRUGD *has ratio at least*

$$\alpha_2(\Delta) = \frac{1}{2} + \frac{7\Delta - 9}{6\Delta(3\Delta - 2)}.$$

Definition 5.2 (The triple mixture)**.** For $n \geq 6$, define T by

$$T = \frac{5255}{6186}\,\text{Perm} + \frac{840}{6186}\,\text{PRUGD} + \frac{91}{6186}\,D_2.$$

That is, on input G, run Perm with probability 5255/6186, run PRUGD with probability 840/6186, and run D_2 with probability 91/6186.

Theorem 5.3. *For every $n \geq 6$, the triple mixture T is impartial and has performance guarantee at least*

$$\beta_n = \frac{12421}{18558} - \frac{273}{24744\,n}.$$

Consequently,

$$\liminf_{n\to\infty} \inf_{G\in\mathcal{G}_n:\Delta(G)>0} \frac{\mathbb{E}[d_G(T(G))]}{\Delta(G)} \geq \frac{12421}{18558} = 0.669307\ldots.$$

Proof. Impartiality follows because T is a convex combination of impartial mechanisms.

Let

$$A = \frac{5255}{6186}, \qquad B = \frac{840}{6186}, \qquad C = \frac{91}{6186}.$$

The weights were chosen so that

$$A \cdot \frac{2}{3} + B \cdot \frac{65}{96} + C \cdot \frac{3}{4} = \frac{12421}{18558}, \tag{1}$$

$$A \cdot \frac{2}{3} + B \cdot \frac{13}{18} + C \cdot \frac{1}{3} = \frac{12421}{18558}, \tag{2}$$

$$A \cdot \frac{31}{45} + B \cdot \frac{25}{42} + C \cdot \frac{2}{9} = \frac{12421}{18558}. \tag{3}$$

If $\Delta(G) = 2$, Proposition 5.1(i) and Theorem 3.1 give

$$\begin{aligned}\frac{\mathbb{E}[d_G(T(G))]}{\Delta(G)} &\geq A \cdot \frac{2}{3} + B \cdot \frac{65}{96} + C\left(\frac{3}{4} - \frac{3}{4n}\right) \\ &= \frac{12421}{18558} - \frac{273}{24744\,n}.\end{aligned}$$

If $\Delta(G) = 3$ and there is a unique vertex of indegree at least 2, then Proposition 5.1(ii) and Lemma 4.1 imply, using (2),

$$\frac{\mathbb{E}[d_G(T(G))]}{\Delta(G)} \geq \frac{12421}{18558}.$$

If $\Delta(G) = 3$ and at least two vertices have indegree at least 2, then Proposition 5.1(iii) and Lemma 4.2 imply, using (3),

$$\frac{\mathbb{E}[d_G(T(G))]}{\Delta(G)} \geq \frac{12421}{18558}.$$

It remains to consider $\Delta \geq 4$. Since D_2 has nonnegative expected indegree, it suffices to prove

$$A\alpha_1(\Delta) + B\alpha_2(\Delta) \geq \frac{12421}{18558}.$$

For each even $\Delta \geq 4$, the value of $\alpha_1(\Delta)$ is the same as the value of $\alpha_1(\Delta + 1)$, while α_2 is decreasing. Hence it is enough to check odd $\Delta \geq 5$. For odd $d \geq 5$,

$$\alpha_1(d) = \frac{3}{4} - \frac{1}{4d},$$

and therefore

$$A\alpha_1(d) + B\alpha_2(d) = \frac{5(10467d^2 - 9347d + 1094)}{24744d(3d-2)}.$$

The derivative of the right-hand side is

$$\frac{5(7107d^2 - 6564d + 2188)}{24744d^2(3d-2)^2},$$

which is positive for all real d because the quadratic in the numerator has negative discriminant. Thus the minimum over odd $d \geq 5$ is attained at $d = 5$, where the value is

$$A\alpha_1(5) + B\alpha_2(5) = \frac{8309}{12372} = \frac{12421}{18558} + \frac{85}{37116} > \frac{12421}{18558}.$$

This completes the case analysis. □

Corollary 5.4. *The asymptotic lower bound in Theorem 5.3 exceeds* $2105/3147$ *by*

$$\frac{12421}{18558} - \frac{2105}{3147} = \frac{8099}{19467342} = 0.000416\ldots.$$

Moreover, $\beta_n > 2105/3147$ *for every* $n \geq 27$.

Proof. The displayed difference is a direct calculation. The finite-n statement follows from

$$\frac{273}{24744n} < \frac{8099}{19467342} \iff n > \frac{9441}{356},$$

which holds for all integers $n \geq 27$. □

6 What remains open

The mechanism D_2 removes the most visible obstruction coming from graphs of maximum indegree 2: on that degree class it reaches $3/4 - O(1/n)$. The full asymptotic 3/4 problem remains open after the arguments in this note. The reason is that D_2 is deliberately narrow: it assigns probability only to vertices that have indegree 2 after the default edge is removed. It therefore contributes little on graphs of larger maximum indegree, and only a small mixing weight can be used without damaging the guarantees supplied by the permutation mechanism and PRUGD.

A plausible next target is the fixed-degree regime $\Delta = 3$. Lemma 4.2 is intentionally crude; any stronger impartial primitive for graphs with one or more indegree-two runner-up vertices would directly improve the mixture calculation. More ambitiously, a mechanism that treats maximum and runner-up vertices in a balanced leave-one-out way for every fixed Δ could be a route toward the conjectural 3/4 limit.

References

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